TrustChain-AI: A Decentralized Trust Intelligence Framework for Secure Autonomous AI Agent Collaboration

A Project Report

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CERTIFICATE

This is to certify that the project titled "TrustChain-AI: A Decentralized Trust Intelligence Framework for Secure Autonomous AI Agent Collaboration" has been submitted by the following students in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology in Computer Engineering of NMIMS University, Mumbai during the academic year 2026–2027.

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The project was carried out under our guidance and supervision. To the best of our knowledge and belief, the work presented in this report is genuine and has not been submitted for any other degree, diploma, or award.

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ABSTRACT

The rapid proliferation of autonomous AI agent systems has created an urgent need for robust mechanisms to establish, monitor, and enforce trust among collaborating agents. Existing multi-agent platforms lack cryptographically verifiable identity primitives, continuous behavioural trust monitoring, and dynamic authorization policies that respond to runtime agent behaviour. Without these foundations, AI agent systems are vulnerable to impersonation, credential forgery, policy bypass, and Byzantine agent behaviour.

This project presents TrustChain-AI, a decentralised trust intelligence framework for secure autonomous AI agent collaboration. The framework integrates W3C Decentralised Identifiers (DID:key, Ed25519) and Verifiable Credentials (Ed25519Signature2020, JSON-LD canonicalisation) to establish cryptographically verifiable agent identity. A temporal Exponential Moving Average (EMA) trust scoring engine continuously evaluates agent behaviour across task success, policy compliance, and response latency, transitioning agents among TRUSTED, SUSPICIOUS, QUARANTINED, and REVOKED states. An adaptive policy enforcement module gates operations based on live trust state.

Implemented as a FastAPI (Python 3.12.3) backend with a 14-table SQLite local Verifiable Data Registry and a React 19 / Vite 8 frontend dashboard, the system exposes 39 REST endpoints secured by JWT authentication and role-based access control. A complete test suite of 77 automated tests passes without failures. All six stated project objectives are verified as achieved.

Keywords: Decentralised Identifiers, Verifiable Credentials, Multi-Agent Systems, Trust Management, EMA, Adaptive Authorization, Zero Trust, Self-Sovereign Identity

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LIST OF ABBREVIATIONS AND SYMBOLS

Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
ASGI	Asynchronous Server Gateway Interface
DID	Decentralised Identifier
EMA	Exponential Moving Average
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
IoT	Internet of Things
JSON	JavaScript Object Notation
JSON-LD	JavaScript Object Notation for Linked Data
JTI	JWT Token Identifier
JWT	JSON Web Token
MAS	Multi-Agent System
MPSTME	Mukesh Patel School of Technology Management & Engineering
NMIMS	SVKM's Narsee Monjee Institute of Management Studies
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDK	Software Development Kit
SOC	Security Operations Centre
SQLite	Self-Contained SQL Database Engine
SSI	Self-Sovereign Identity

URL	Uniform Resource Locator
UUID	Universally Unique Identifier
VDR	Verifiable Data Registry
VC	Verifiable Credential
W3C	World Wide Web Consortium
WAL	Write-Ahead Logging
ZTA	Zero Trust Architecture

Symbols and Notations

Symbol	Definition
P_i(t)	Multi-factor performance score for agent i at time t
T_i(t)	Temporal trust score (EMA) for agent i at time t
S_i(t)	Task success rate for agent i over evaluation window
C_i(t)	Policy compliance rate for agent i over evaluation window
L_i(t)	Normalised response latency for agent i
w1 = 0.6	Weight for task success factor
w2 = 0.3	Weight for policy compliance factor
w3 = 0.1	Weight for latency factor (inverted)
lambda = 0.7	EMA decay factor
tau_T = 0.8	TRUSTED state threshold
tau_S = 0.5	SUSPICIOUS state threshold
n = 10	Evaluation window size (telemetry events)

CHAPTER 1: INTRODUCTION

1.1 Background and Motivation

The deployment of autonomous AI agents has accelerated dramatically across enterprise and research settings. AI agents perform data analysis, threat detection, code generation, document summarisation, and process automation at machine speed, often with minimal human supervision. In multi-agent orchestration platforms, one AI agent may delegate subtasks to another, consuming the output of upstream agents as trusted inputs. This introduces a fundamental security question: how does the system verify that the agent it is trusting is authentic, behaving correctly, and authorised for the requested operation?

Traditional security mechanisms — API keys, OAuth tokens, TLS certificates — address authentication but not continuous behavioural trust. An agent holding a valid credential may produce incorrect results due to model drift, adversarial manipulation, or internal misconfiguration. Static role-based access control grants permissions at registration and cannot revoke them in response to runtime behavioural anomalies. In collaborative AI pipelines, a single compromised agent can corrupt the entire workflow chain.

The academic field of trust and reputation management in distributed systems (Josang et al. [3], EigenTrust [4], Mui et al. [5]) provides formal frameworks for trust quantification from interaction history, but these were designed for peer networks of homogeneous nodes, not heterogeneous AI agents with diverse capability profiles and cryptographic identity requirements. The W3C Decentralised Identifiers (DID) v1.0 [1] and Verifiable Credentials (VC) v1.1 [2] standards provide a foundation for self-sovereign digital identity. NIST Special Publication 800-207 [15] articulates Zero Trust Architecture (ZTA) — continuous verification of identity and posture for every resource access.

TrustChain-AI integrates these intellectual strands — trust and reputation management, self-sovereign identity, and zero trust architecture — into a coherent, implementable, and testable framework specifically designed for secure autonomous AI agent collaboration.

1.2 Problem Statement

Existing multi-agent AI platforms lack a unified trust management framework that combines:

- (a) Cryptographically verifiable agent identity — existing API keys can be forged or shared without cryptographic detection.
- (b) Continuous behavioural trust monitoring — agent authorisation is static; runtime degradation is undetected.
- (c) Trust-state-aware adaptive authorisation — no mechanism gates high-privilege operations based on live trust state.
- (d) Auditable trust evidence — no persistent, structured record of the evidence supporting each trust decision.
- (e) Trust-weighted capability matching — agent selection ignores reliability and trust history.

These gaps expose multi-agent AI systems to impersonation, Byzantine agent behaviour, privilege escalation, and credential forgery threats that cannot be mitigated by authentication mechanisms alone.

1.3 Project Objectives

1. Objective 1: Implement W3C DID:key method with Ed25519 for cryptographically verifiable agent identity.
2. Objective 2: Implement W3C Verifiable Credentials (Ed25519Signature2020, JSON-LD) for agent credential issuance, storage, and verification.
3. Objective 3: Design and implement a temporal EMA trust scoring engine evaluating success, compliance, and latency.
4. Objective 4: Implement adaptive policy enforcement gating agent operations based on live trust state.
5. Objective 5: Build a trust-weighted agent discovery and matching engine with multi-factor composite scoring.
6. Objective 6: Implement sequential multi-agent workflow orchestration with per-step trust verification.

1.4 Scope and Boundaries

In Scope

- All six objectives listed in Section 1.3.
- Local Verifiable Data Registry implemented as a SQLite database (vdr.db) with Write-Ahead Logging.
- Structured local execution engine (LocalAgentConnector) for agent task simulation.
- JWT authentication: Argon2id hashing, access/refresh tokens with rotation, account lockout.
- RBAC with three roles: ADMIN, OPERATOR, VIEWER.
- React 19 / Vite 8 dashboard with 17 interactive components.
- Complete automated test suite of 77 tests.

Out of Scope

- Distributed or blockchain-based VDR (future scope — Section 6.3.1).
- Live LLM inference for agent task execution (future scope — Section 6.3.2).
- PARALLEL workflow execution (schema-supported but not tested — Section 6.3.3).
- Cross-organisational federated trust (future scope — Section 6.3.4).

1.5 Proposed Solution Overview

TrustChain-AI is organised into five architectural layers: (1) Presentation Layer — React 19 SPA with 17 components; (2) API Gateway Layer — FastAPI 0.141.1, Pydantic v2 validation, JWT RBAC, OpenTelemetry; (3) Trust Intelligence Layer — TrustEngine EMA scoring, AdaptivePolicyEnforcer, LocalAgentConnector; (4) Identity and Credential Layer — DIDKey Ed25519, VerifiableCredential Ed25519Signature2020; (5) Persistence Layer — SQLite WAL VDR, SQLAlchemy 2.0, 14 tables.

[DIAGRAM: Overall System Architecture]

See docs/blackbook/diagrams/all_diagrams.md — Figure 1

Figure 1: Overall System Architecture

1.6 Key Contributions

- Novel integrated framework combining W3C DID/VC standards with EMA behavioural trust scoring.
- Adaptive policy enforcement operationalising Zero Trust principles for AI agent operations.
- Trust-aware multi-factor agent discovery algorithm (trust 0.4, reputation 0.3, success 0.2, latency 0.1).
- Complete four-dimension audit trail: trust transitions, security events, collaboration events, VC records.

1.7 Report Organisation

Chapter 2 surveys trust management, SSI, zero trust, and multi-agent systems literature. Chapter 3 presents the complete system design and implementation. Chapter 4 reports verified test results and system evidence. Chapter 5 discusses advantages, limitations, and applications. Chapter 6 concludes and identifies future scope. Appendices A–H provide the API reference, DB schema, test results, trust mathematics, algorithms, dependencies, source excerpts, and QA checklist.

1.8 Chapter Summary

This chapter established the motivation for a unified trust management framework for multi-agent AI systems, identified five specific gaps in existing platforms, defined six concrete project objectives, and described the scope and structure of TrustChain-AI.

CHAPTER 2: LITERATURE SURVEY

2.1 Introduction

This chapter surveys fifteen works across five thematic areas: (1) trust and reputation in distributed systems, (2) trust in IoT and cyber-physical systems, (3) decentralised identity and SSI, (4) adaptive authorization and zero trust, and (5) multi-agent systems coordination. A research gap analysis identifies the specific gaps that TrustChain-AI addresses.

2.2 Trust and Reputation in Distributed Systems

Josang, Ismail, and Boyd [3] published a comprehensive survey of trust and reputation systems, distinguishing reliability trust, competence trust, and honesty trust, and classifying systems by aggregation method, evidence type, and trust model. The survey remains the primary reference taxonomy for evaluating new trust systems.

Kamvar, Schlosser, and Garcia-Molina [4] proposed EigenTrust for P2P reputation management: a normalised global trust value computed by iterative propagation from pre-trusted peers, converging to the principal eigenvector of the trust matrix. While attack-resistant, EigenTrust assumes a static graph and provides a single scalar trust value without distinguishing success, compliance, or latency dimensions — insufficient for AI agent heterogeneity.

Mui, Mohtashemi, and Halberstadt [5] proposed a Bayesian computational trust model using a Beta distribution over cooperative/non-cooperative outcomes. Mathematically rigorous but limited to binary cooperation; multi-dimensional agent quality is not captured.

Azzedin and Maheswaran [7] studied context-sensitive, hierarchical trust for grid computing. Their concept of task-type-specific trust influenced TrustChain-AI's capability-specific match scoring.

2.3 Trust in IoT and Cyber-Physical Systems

Nitti, Girau, and Atzori [6] studied trust in the Social IoT, distinguishing object trust, service trust, and social relationship trust. Their multi-dimensional, temporal trust

approach directly parallels TrustChain-AI's three-factor EMA model (S, C, L). Gap: IoT devices lack cryptographic self-sovereign identity and graduated authorisation enforcement.

2.4 Decentralised Identity and Self-Sovereign Identity

W3C DID Core v1.0 [1] defines a URI scheme for globally unique, cryptographically verifiable decentralised identifiers. The did:key method generates a DID from a public key with no external infrastructure. W3C VC Data Model v1.1 [2] defines digitally-signed credential format with issuer, credentialSubject, and proof blocks.

RFC 8037 [8] specifies Ed25519/X25519 in the JOSE framework. Fromknecht et al. [9] proposed CertCoin — a decentralised PKI using blockchain for identity-to-key binding. The Trust over IP Foundation [14] defines a four-layer decentralised trust model. TrustChain-AI addresses ToIP layers 2–4 (DID communication, credential exchange, application ecosystem) while explicitly excluding Layer 1 (blockchain infrastructure) from scope.

2.5 Adaptive Authorization and Zero Trust

NIST SP 800-207 [15] defines seven Zero Trust tenets. TrustChain-AI operationalises tenets 4 (dynamic policy), 6 (dynamic authorisation), and 7 (continuous monitoring) for AI agents. Zhang et al. [11] proposed decentralised trust-based access control for smart data, analogous to TrustChain-AI's operation-level trust-state gating. Bernabe et al. [10] surveyed privacy-preserving blockchain access control, highlighting the auditability–privacy tension relevant to TrustChain-AI's audit trail design.

2.6 Multi-Agent Systems and Agent Coordination

FIPA [12] defines the Agent Management System (AMS) as a white pages registry for agent identity and lifecycle management. TrustChain-AI's agent registry extends the FIPA AMS model with trust-state management, capability-based yellow pages discovery, and adaptive authorisation enforcement. Shoker [13] proposed a self-stabilising autonomous distributed ledger relevant as a theoretical reference for the distributed VDR future scope.

2.7 Research Gap Analysis

Table 11: Research Gap Analysis Matrix

Prior Work	Focus	Key Limitation	TrustChain-AI Contribution
Josang et al. [3]	Trust taxonomy	No cryptographic identity; single-dimension	Multi-factor EMA + DID/VC identity
EigenTrust [4]	P2P reputation	Static graph; scalar trust	Dynamic registry; 4-state machine; VC identity
Mui et al. [5]	Bayesian trust	Binary metric; no capability dimension	3-factor performance score; capability matching
Nitti et al. [6]	IoT trust	No crypto identity; no adaptive auth	DID:key identity; adaptive policy enforcement
W3C DID/VC [1][2]	Decentralised identity	Human identity focus; no behavioural trust	AI agent identity + EMA trust coupling
NIST ZTA [15]	Zero trust model	Organisational policy; not agent-specific	Operation classification for AI agent domain
FIPA AMS [12]	Agent lifecycle mgmt	No crypto identity; no trust scoring	DID registry + trust scoring + adaptive authz

2.8 Chapter Summary

No prior work integrates all five dimensions: cryptographic verifiable identity, multi-factor behavioural trust scoring, trust-state-aware adaptive authorisation, capability-based trust-weighted discovery, and auditable evidence recording. TrustChain-AI addresses all five in a single testable implementation.

CHAPTER 3: METHODOLOGY AND IMPLEMENTATION

3.1 Introduction

This chapter presents the complete technical design and implementation of TrustChain-AI. It covers the five-layer architecture, identity management, trust intelligence engine, adaptive authorisation, agent registry and discovery, collaboration, workflow orchestration, security architecture, database design, REST API, and frontend. All implementation details are derived from the verified source code.

3.2 System Architecture

3.2.1 Five-Layer Architecture

TrustChain-AI is organised into five logical layers (Figure 2). Layer 1 (Presentation) provides a React 19 SPA with 17 interactive components, Recharts visualisations, and dark/light theme toggle. Layer 2 (API Gateway) provides FastAPI with Pydantic v2 validation, JWT RBAC, and OpenTelemetry. Layer 3 (Trust Intelligence) implements TrustEngine EMA scoring, AdaptivePolicyEnforcer, and LocalAgentConnector. Layer 4 (Identity and Credential) provides DIDKey Ed25519 and VerifiableCredential Ed25519Signature2020. Layer 5 (Persistence) provides SQLite WAL VDR with SQLAlchemy 2.0 and 14 tables.

[DIAGRAM: Five-Layer Architecture]

See docs/blackbook/diagrams/all_diagrams.md — Figure 2

Figure 2: Five-Layer Architecture

3.2.2 Technology Stack

Table 6: Technology Stack Summary

Component	Technology	Version	Purpose
Web Framework	FastAPI	0.141.1	REST API, routing, dependency injection
ASGI Server	Uvicorn	0.52.4	Async server
ORM	SQLAlchemy	2.0.52	Database abstraction

Validation	Pydantic	2.13.5	Request/response schema
Cryptography	PyNaCl	1.6.2	Ed25519 keypair and signing
JWT	PyJWT	2.14.0	Access and refresh tokens
JSON-LD	PyLD	3.2.0	VC canonicalisation
Password	argon2-cffi	25.1.0	Argon2id hashing
Base58	base58	2.1.1	DID multibase encoding
HTTP Client	httpx	0.28.1	Agent endpoint calls
Observability	opentelemetry	1.44.0	Request instrumentation
Runtime	Python	3.12.3	Backend runtime
Frontend	React + Vite	19.2.8 + 8.2.2	SPA dashboard
Routing (FE)	React Router DOM	7.18.3	Client-side navigation
Charts	Recharts	3.10.1	Trust analytics visualisation
Styling	Tailwind CSS	4.3.3	Utility-first CSS
API Client	Axios	1.20.0	HTTP with JWT interceptor

3.3 Identity Management

3.3.1 DID:key Generation (akta/did.py — 452 lines)

The DIDKey class implements the W3C DID:key method with Ed25519 cryptography. The generation process (Figure 6):

7. Step 1: Generate Ed25519 keypair (32-byte private key, 32-byte public key) via PyNaCl.
8. Step 2: Prepend multicodec prefix [0xED, 0x01] to produce 34-byte prefixed key.

9. Step 3: Base58btc-encode with multibase prefix 'z' → multibase string.
10. Step 4: Construct DID URI: did:key:z{multibase_string}.
11. Step 5: Build DID Document with Ed25519VerificationKey2020 verification method.

Algorithm 1: DID:key Generation

Algorithm 1 — DID:key Generation (from akta/did.py)

```
Input:  (none — generates fresh keypair)
Output: did_uri (string), did_document (JSON-LD), private_key (bytes)

1. signing_key  ← Ed25519.generate()          // PyNaCl
2. public_key   ← signing_key.verify_key.encode()  // 32 bytes
3. prefixed     ← [0xED, 0x01] + public_key   // multicodec Ed25519
4. multibase    ← 'z' + base58btc(prefixed)
5. did_uri      ← 'did:key:' + multibase
6. vm_id        ← did_uri + '#' + multibase
7. did_document ← {
     '@context': [DID_CONTEXT, ED25519_CONTEXT],
     'id': did_uri,
     'verificationMethod': [{
       'id': vm_id, 'type': 'Ed25519VerificationKey2020',
       'controller': did_uri, 'publicKeyMultibase': multibase
     }],
     'authentication':        [vm_id],
     'assertionMethod':       [vm_id],
     'capabilityInvocation':  [vm_id],
     'capabilityDelegation':  [vm_id]
   }
8. return did_uri, did_document, signing_key.encode()
```

3.3.2 Verifiable Credentials (akta/credentials.py)

The VerifiableCredential class issues W3C VCs with Ed25519Signature2020 proofs. The signing process: (1) construct credential JSON-LD without proof block; (2) canonicalise using URDNA2015 via PyLD 3.2.0; (3) sign canonical bytes with issuer Ed25519 private key via PyNaCl; (4) base58-encode 64-byte signature with 'z' prefix; (5) embed as proofValue in Ed25519Signature2020 proof block. VCs are stored in the stored_vcs table.

[DIAGRAM: Verifiable Credential Issuance and Verification Sequence]

See docs/blackbook/diagrams/all_diagrams.md — Figure 7

Figure 7: Verifiable Credential Issuance and Verification Sequence

3.4 Trust Intelligence Engine

3.4.1 Multi-Factor Performance Score

The trust scoring engine (akta/trust_engine.py, akta/vdr/trust_crud.py) evaluates three independent behavioural dimensions over a sliding window of the last n = 10 telemetry events:

S_i(t): task success rate — proportion of events where success = True.

C_i(t): policy compliance rate — proportion of events where compliance_event = True.

L_i(t): normalised latency = min(avg_latency_ms / 2000, 1.0).

The three factors are combined into a composite performance score:

$$P_i(t) = 0.6 * S_i(t) + 0.3 * C_i(t) + 0.1 * (1 - L_i(t)) \quad \text{... (Eq. 1)}$$

Table 1: Trust Weight Configuration

Factor	Symbol	Weight	Rationale
Task success rate	S_i(t)	w1 = 0.6	Primary obligation — task completion
Policy compliance rate	C_i(t)	w2 = 0.3	Security signal — policy adherence
Normalised latency (inv.)	1 - L_i(t)	w3 = 0.1	Performance quality signal
Total		1.0	Ensures P_i(t) in [0,1]

3.4.2 Temporal Trust Score (EMA)

The performance score is temporally smoothed using an Exponential Moving Average with decay factor lambda = 0.7, weighting recent behaviour more heavily while retaining historical context:

$$T_i(t) = 0.7 * P_i(t) + 0.3 * T_i(t-1) \quad \text{... (Eq. 2)}$$

Initial trust T_i(0) = 1.0 (TRUSTED) — benefit of the doubt for newly registered agents. The REVOKED state is terminal; telemetry submissions cannot update a REVOKED agent's score.

3.4.3 Trust State Machine

Table 2: Trust State Thresholds and Transitions

Trust State	Condition	Meaning	Recovery Path
TRUSTED	T >= 0.8	Agent performing reliably and compliantly	—
SUSPICIOUS	0.5 <= T < 0.8	Behaviour degraded; heightened monitoring	Automatic via telemetry
QUARANTINED	T < 0.5	Severely degraded; collaboration restricted	Automatic via telemetry
REVOKED	Admin command	Permanently revoked; all access blocked	Manual re-registration only

[DIAGRAM: Trust State Transition Diagram]

See docs/blackbook/diagrams/all_diagrams.md — Figure 4

Figure 9: Trust State Transition Diagram

3.4.4 Trust Evaluation Algorithm

Algorithm 2 — Trust Evaluation (from akta/vdr/trust_crud.py)

```
Input:  agent_did, telemetry_event {success, compliance_event, latency_ms}
Output: updated trust_score, trust_state

1.  Insert telemetry_event into telemetry_logs for agent_did
2.  window ← last 10 telemetry_logs records for agent_did (by timestamp
DESC)
3.  n     ← len(window)
4.  S_i   ← count(success=True in window) / n
5.  C_i   ← count(compliance_event=True in window) / n
6.  avg_L ← mean(latency_ms for events in window)
7.  L_i   ← min(avg_L / 2000.0, 1.0)
8.  P_i   ← 0.6*S_i + 0.3*C_i + 0.1*(1 - L_i)
9.  T_prev ← agent_trust_states.temporal_trust_score for agent_did (default
1.0)
10. T_i   ← 0.7*P_i + 0.3*T_prev
11. if   T_i >= 0.8: state ← TRUSTED
    elif T_i >= 0.5: state ← SUSPICIOUS
```

```
    else:            state ← QUARANTINED
12. UPDATE agent_trust_states SET temporal_trust_score=T_i,
trust_state=state
13. INSERT trust_audit_trails (prev_state, new_state, prev_score,
new_score, reason)
14. return T_i, state
```

[DIAGRAM: Trust Evaluation Flowchart]

See docs/blackbook/diagrams/all_diagrams.md — Figure 3

Figure 8: Trust Evaluation Flowchart

3.5 Adaptive Authorization

3.5.1 Operation Classification (akta/adaptive_auth.py — 211 lines)

PRIVILEGED_OPERATIONS (denied to SUSPICIOUS agents):

```
workflow:execute, workflow:orchestrate, agent:delegate,
credential:issue, admin:revoke, admin:manage, sensitive:execute
```

STANDARD_OPERATIONS (permitted to SUSPICIOUS agents with audit):

```
read, query, map:generate, task:execute, collaboration:step,
collaboration:participate
```

3.5.2 Authorization Matrix

Table 3: Adaptive Authorization Matrix

Trust State	Privileged Operations	Standard Operations	Audit Recorded
TRUSTED	Permitted	Permitted	On denial only
SUSPICIOUS	HTTP 403	Permitted	Always
QUARANTINED	HTTP 403	HTTP 403	Always
REVOKED	HTTP 403	HTTP 403	Always

[DIAGRAM: Adaptive Authorization Flow]

See docs/blackbook/diagrams/all_diagrams.md — Figure 5

Figure 10: Adaptive Authorization Flow

3.6 Agent Registry and Discovery

3.6.1 Agent Registry (akta/vdr/registry_crud.py — 733 lines)

The registry stores agent records in the agents table. Lifecycle: Registration (DID, capabilities, initial trust 1.0/TRUSTED) → Telemetry Submission (triggers trust computation) → Health Tracking (healthy/degraded/offline endpoint status) → Ratings (peer scores 1.0–5.0 averaging to reputation_score).

3.6.2 Trust-Weighted Discovery

The match score formula for agent i:

$$\texttt{match_score_i} = 0.4 \cdot T_i + 0.3 \cdot (r_i / 5.0) + 0.2 \cdot \sigma_i + 0.1 \cdot (1 - \lambda_i) \quad \text{... (Eq. 3)}$$

where T_i = temporal trust score, r_i = reputation score (1–5), σ_i = successful_tasks/total_tasks (default 1.0), λ_i = min(last_response_ms/2000, 1.0).

Table 4: Agent Match Scoring Weights

Factor	Symbol	Weight
Temporal trust score	T_i	0.4
Normalised reputation	$r_i / 5.0$	0.3
Task success rate	σ_i	0.2
Normalised latency (inv.)	$1 - \lambda_i$	0.1

3.7 Multi-Agent Collaboration

3.7.1 Collaboration Session Lifecycle

Sessions are initiated via POST /registry/collaborate. Both agents' trust states are verified (QUARANTINED/REVOKED agents cannot participate). Sessions proceed: creation (status=active) → message exchange (task_request → response via LocalAgentConnector) → completion (productivity scoring, automatic telemetry for both agents).

3.7.2 LocalAgentConnector (akta/connectors.py — 135 lines)

IMPORTANT: The LocalAgentConnector is a structured local execution engine, NOT live LLM inference. It produces capability-specific structured text responses (research summary, analysis report, pseudocode, step-plan, verification checklist) based on the agent's declared capability. For agents with HTTP endpoints starting with 'http', a real HTTP POST is attempted; local simulation is used as fallback. Live LLM integration is identified as future scope (Section 6.3.2).

3.8 Workflow Orchestration

3.8.1 Sequential Workflow Pipeline

Workflows are created with a name, description, SEQUENTIAL pattern, and ordered steps (each specifying required_capability). Execution: (1) update workflow status to running; (2) for each node in step_order: discover highest-match TRUSTED agent with required capability → verify trust state → execute via LocalAgentConnector with previous node's output_data as input → update node status and output_data → submit telemetry for executing agent; (3) update workflow to completed.

[DIAGRAM: Sequential Workflow Execution Diagram]

See docs/blackbook/diagrams/all_diagrams.md — Figure 7

Figure 13: Sequential Workflow Execution Diagram

3.8.2 PARALLEL Pattern Status

The workflows table includes a pattern column accepting SEQUENTIAL and PARALLEL. The execution engine implements only SEQUENTIAL. PARALLEL workflow execution is schema-defined but not implemented; it is identified as future scope (Section 6.3.3).

3.9 Security Architecture

3.9.1 Authentication (akta/auth/)

- Password Hashing: Argon2id via argon2-cffi 25.1.0 (OWASP recommended).
- JWT Access Tokens: 30-minute TTL, HS256, payload includes sub, role, jti, exp.

- JWT Refresh Tokens: 7-day TTL with rotation — each token single-use, SHA-256 hash stored in DB.
- Reuse Detection: Revoked token presented → all user tokens revoked, HTTP 401 returned.
- Account Lockout: 5 consecutive failures → locked_until timestamp set, HTTP 423 on further attempts.

3.9.2 RBAC Roles

ADMIN: Full access including user management, agent revocation, security audit events. OPERATOR: Agent registration, telemetry, collaboration, workflow creation and execution. VIEWER: Read-only access to all data.

3.9.3 Cryptographic Layer

Ed25519 keypairs via PyNaCl 1.6.2 (libsodium). Ed25519Signature2020 VC proofs. JSON-LD URDNA2015 canonicalisation via PyLD 3.2.0. Base58btc multibase via base58 2.1.1. JWT HS256 via PyJWT 2.14.0.

3.10 Database Design

The local SQLite VDR (vdr.db) uses WAL mode and SQLAlchemy 2.0 ORM (models in akta/vdr/models.py, 194 lines). It is NOT a distributed ledger or blockchain; it is a local SQLite-based VDR implementation suitable for single-organisation development and demonstration.

Table 5: Database Table Summary (13 core tables — agent_trust_states listed once)

Table	Group	Key Columns	Purpose
agent_trust_states	Trust	agent_did, temporal_trust_score, trust_state	Current trust per agent
telemetry_logs	Trust	agent_did, success, compliance_event, latency_ms	Raw telemetry events
trust_audit_trails	Trust	agent_did, previous_state,	State transition audit

		new_state, reason	
stored_vcs	Identity	vc_id, issuer_did, subject_did, credential_data	VC storage
agents	Registry	agent_id, did, capabilities, reputation_score	Agent registry
agent_ratings	Registry	agent_id, rating, feedback	Peer ratings
collaboration_sessions	Collaboration	session_id, requester_did, target_did, status	Session records
collaboration_events	Collaboration	session_id, sender_did, message_type, content	Message log
workflows	Workflow	workflow_id, pattern, status, result	Workflow definitions
workflow_nodes	Workflow	workflow_id, agent_did, step_order, output_data	Workflow steps
users	Auth	user_id, username, role, password_hash	Operator accounts
refresh_tokens	Auth	jti, token_hash, user_id, revoked_at	JWT refresh store
security_audit_events	Auth	event_type, user_id, ip_address, timestamp	Security event log

[DIAGRAM: Database Entity-Relationship Diagram]

See docs/blackbook/diagrams/all_diagrams.md — Figure 9

Figure 14: Database Entity-Relationship Diagram

3.11 REST API Design

Table 7: API Endpoint Summary by Module

Module	Base Path	Count	Key Operations
Authentication	/auth	9	Register, login, refresh, logout, user mgmt, audit events
Agent Registry	/registry	21	Agent CRUD, telemetry, trust, discover, auth, collaborate, workflow
VDR / Credentials	/vdr	7	VC issuance/retrieval, DID generate/resolve/verify
Core / Health	/, /health	2	Root, system health probe

3.12 Frontend Architecture

17 React components in frontend/src/components/: DashboardHome (KPI cards, trend charts), AgentExplorer (registry table, trust badges), AgentDetails (trust history, telemetry log), DiscoveryMatching (capability search, ranked results), TrustAnalytics (Recharts visualisations), TelemetryCenter (submission UI), StateTransitionMap (state machine view), AuthorizationVisualizer (policy matrix), CollaborationCenter (session management), WorkflowCenter (workflow creation and execution), IdentityCredentials (DID generation, VC issuance), AuditTrail (event log browser), SystemHealth (/health probe display), UserManagement (RBAC admin), DemoVisualizer (guided lifecycle demo), LoginPage, TopBar (theme toggle, logout).

State management: Axios 1.20.0 with JWT Bearer interceptor; React Router DOM 7.18.3 SPA routing; ThemeContext + CSS custom properties for Dark/Light theme (persisted to localStorage); Recharts 3.10.1 for trust analytics charts.

3.13 Development Setup and Dependencies

Development Environment Commands

```
# Backend startup
.venv\Scripts\python.exe -m uvicorn akta.server:app --host 0.0.0.0 --port
```

```
8050

# Frontend startup (from frontend/ directory)
npm run dev   # starts Vite dev server on http://localhost:5173

# Run test suite
.venv\Scripts\python.exe -m pytest tests/ -v

# Python runtime: 3.12.3
# Node.js: required for frontend (npm)
```

3.14 Chapter Summary

This chapter presented the complete design and implementation of TrustChain-AI across five layers. The identity layer implements W3C DID:key and Ed25519Signature2020 VCs. The trust intelligence layer implements three-factor EMA scoring with a four-state machine. The adaptive authorisation layer combines RBAC with trust-state gating. The 14-table SQLite VDR provides full audit persistence. The React frontend provides complete operator visibility.

CHAPTER 4: RESULTS AND ANALYSIS

4.1 Introduction

This chapter presents verification evidence for all six project objectives based on three evidence categories: (1) automated test suite results — 77/77 tests passing (run 2026-10-16, Python 3.12.3); (2) live system health data from the running backend; (3) mathematical verification through worked examples.

4.2 Test Suite Results

Table 8: Test Suite Results Summary — 2026-10-16, Python 3.12.3, 11.37s

Test File	Tests	Coverage Area	Result
tests/commands/ (CLI tests)	6	CLI command structure	6/6 PASSED
test_adaptive_policy.py	8	All trust state × operation type combinations	8/8 PASSED
test_authentication.py	19	JWT, Argon2id, rotation, lockout, RBAC, audit	19/19 PASSED
test_cli.py	5	CLI command execution	5/5 PASSED
test_e2e_lifecycle.py	3	End-to-end: success, degradation, recovery/revocation	3/3 PASSED
test_multi_agent.py	3	Multi-agent registry, discovery, collaboration, quarantine	3/3 PASSED
test_platform_features.py	3	Health monitoring, workflow, trust-aware matching	3/3 PASSED
test_system_health.py	4	/health: healthy, DB	4/4 PASSED

		fail, engine fail, simultaneous fail	
TOTAL	77	All objectives and components	77/77 PASSED

4.2.1 Key Test Descriptions

▶ test_e2e_full_agent_lifecycle_success_path

Registers agent, submits 10 successful compliant low-latency events, verifies TRUSTED state and authorised operations. Validates Objectives 1, 3, 4, 5.

▶ test_e2e_telemetry_degradation_and_quarantine_path

Submits failed, non-compliant, high-latency telemetry; verifies state progression TRUSTED → SUSPICIOUS → QUARANTINED with correct HTTP 403 enforcement. Validates Objectives 3 and 4.

▶ test_e2e_recovery_and_administrative_revocation_path

Demonstrates trust recovery QUARANTINED → SUSPICIOUS → TRUSTED via good telemetry; then verifies admin revocation to REVOKED terminal state. Validates Objectives 3 and 4.

▶ test_feature2_multi_agent_workflow_orchestrator

Creates 3-step SEQUENTIAL workflow, executes, verifies each node transitions pending → running → completed with output_data chaining. Validates Objective 6.

▶ test_feature3_trust_aware_agent_matching

Registers multiple agents with varying trust and reputation; verifies discovery returns agents ranked by descending match score. Validates Objective 5.

▶ test_system_health_healthy

Verifies /health returns status=ok for all three components (FastAPI, SQLite VDR, Trust Math Core).

4.3 System Health Verification

Live /health probe captured 2026-10-16T09:10:00Z (backend running on port 8050):

Live /health Response — all components OK, SQLite latency 4.37 ms


```
{
  "status": "ok",
  "app_name": "Akta Test Agent",
  "version": "1.0.0",
  "components": {
    "fastapi":      { "status": "ok", "port": 8050, "version": "1.0.0" },
    "database":     { "status": "ok", "engine": "sqlite", "mode": "WAL",
                      "latency_ms": 4.37 },
    "trust_engine": { "status": "ok", "mode": "in-process",
                      "weights": { "w1_success": 0.6, "w2_compliance": 0.3,
                                   "w3_latency": 0.1, "decay_lambda": 0.7 }
}
  }
}
```


All three components report status = 'ok'. SQLite VDR probe latency: 4.37 ms. Trust engine reports exact weight configuration matching Section 3.4.1.

4.4 Trust Mathematics Verification

Table 9: Trust Mathematics Worked Examples

Scenario	S_i	C_i	avg L (ms)	L_i	P_i(t)	T_prev	T_i(t)	State
Healthy (10/10 success, 50ms)	1.00	1.00	50	0.025	0.9975	0.90	0.9683	TRUSTED
Degraded (2/10 success, 1500ms)	0.20	0.30	1500	0.750	0.235	1.00	0.4645	QUARANTINED
Recovering (8/10 success, 200ms)	0.80	0.90	200	0.100	0.830	0.50	0.731	SUSPICIOUS

Degraded calculation detail:

```
P_i = 0.6×0.20 + 0.3×0.30 + 0.1×(1−0.750) =
0.120+0.090+0.025 = 0.235
T_i = 0.7×0.235 + 0.3×1.0 = 0.1645+0.300 = 0.4645 →
QUARANTINED (T < 0.5)
```

4.5 Adaptive Authorization Verification

test_adaptive_policy.py (8 tests) — all PASSED:

- [✓] TRUSTED + workflow:execute (PRIVILEGED) → HTTP 200 Permitted
- [✓] TRUSTED + read (STANDARD) → HTTP 200 Permitted
- [✓] SUSPICIOUS + workflow:execute (PRIVILEGED) → HTTP 403 Denied
- [✓] SUSPICIOUS + credential:issue (PRIVILEGED) → HTTP 403 Denied
- [✓] SUSPICIOUS + read (STANDARD) → HTTP 200 Permitted + audit record
- [✓] SUSPICIOUS + task:execute (STANDARD) → HTTP 200 Permitted + audit record
- [✓] QUARANTINED + read (STANDARD) → HTTP 403 Denied
- [✓] REVOKED + read (STANDARD) → HTTP 403 Denied

4.6 Authentication and Security Verification

test_authentication.py (19 tests) — all PASSED. Key verifications:

- Argon2id hash differs from plaintext; correct password verifies True; incorrect verifies False.
- JWT login returns access + refresh token pair with correct claims.
- Refresh token rotation: old token rejected after first use; new token required.
- Reuse detection: presenting revoked token revokes all user tokens; HTTP 401.
- Account lockout: triggered at 5 failed attempts; subsequent attempts return HTTP 423.
- Security audit events recorded for login success, failure, lockout, token revocation.
- Admin can create users and update roles; VIEWER cannot access admin endpoints.

4.7 API Endpoint Verification

All 39 REST endpoints were exercised via pytest TestClient (FastAPI's in-process HTTP test client). All endpoints return correct HTTP status codes and JSON payloads. Complete endpoint listing in Appendix A and docs/blackbook/evidence/api_endpoints.md.

4.8 Frontend Dashboard

The React frontend is operational at http://localhost:5173. All 17 components are confirmed present in frontend/src/components/ and connect to the live backend at port 8050. Dark/light theme toggle functions correctly across all components.

Note: Programmatic browser screenshot capture was not performed during the automated verification run. Screenshots in the screenshots/ directory should be captured manually by the student from the running application for inclusion in the printed report. The following captions indicate where screenshots belong:

[DIAGRAM: Dashboard Home — System Overview]

Capture from http://localhost:5173 — Dashboard Home — System Overview

Figure 19: Dashboard Home — System Overview

[DIAGRAM: Agent Registry and Trust State View]

Capture from http://localhost:5173 — Agent Registry and Trust State View

Figure 20: Agent Registry and Trust State View

[DIAGRAM: Trust Analytics Charts]

Capture from http://localhost:5173 — Trust Analytics Charts

Figure 21: Trust Analytics Charts

[DIAGRAM: Workflow Execution Results]

Capture from http://localhost:5173 — Workflow Execution Results

Figure 22: Workflow Execution Results

[DIAGRAM: System Health Monitoring Screen]

Capture from http://localhost:5173 — System Health Monitoring Screen

Figure 23: System Health Monitoring Screen

4.9 Objectives Achievement Matrix

Table 10: Project Objectives Achievement Matrix — All 6 Objectives Achieved

#	Objective	Implementation	Tests	Status
1	DID:key	akta/did.py (452	test_e2e_lifecycle,	ACHIEVED

	Ed25519 identity	lines)	test_cli	
2	Verifiable Credentials	akta/credentials.py, akta/vdr/router.py	test_e2e_lifecycle	ACHIEVED
3	Temporal EMA Trust Scoring	akta/trust_engine.py, trust_crud.py	test_e2e_lifecycle, test_adaptive_policy	ACHIEVED
4	Adaptive Authorization	akta/adaptive_auth.py (211 lines)	test_adaptive_policy (8 tests)	ACHIEVED
5	Trust-Weighted Discovery	registry_crud.py lines 276–281	test_multi_agent, test_platform_features	ACHIEVED
6	Sequential Workflow	registry_crud.py (733 lines)	test_platform_features (feature2)	ACHIEVED

4.10 Chapter Summary

All 77 automated tests pass. All three backend components report healthy status. Trust math worked examples confirm correct EMA behaviour for degradation, quarantine, and recovery scenarios. The adaptive authorization matrix is verified for all 8 trust state × operation combinations. All 6 project objectives are confirmed achieved.

CHAPTER 5: ADVANTAGES, LIMITATIONS AND APPLICATIONS

5.1 Introduction

This chapter provides an honest assessment of TrustChain-AI's strengths and limitations, followed by an analysis of application domains where the framework's capabilities provide meaningful value.

5.2 Advantages

5.2.1 Cryptographic Identity Assurance

DID:key provides self-certifying, unforgeable identity without a central certificate authority. Possession of the Ed25519 private key can be proved through a digital signature, eliminating the CA single point of trust failure.

5.2.2 Continuous Behavioural Trust Monitoring

EMA-based temporal trust scoring continuously reassesses agent trustworthiness from runtime telemetry. Access rights change dynamically in response to observed behaviour — automatically and without manual intervention.

5.2.3 Adaptive Authorization with Graduated Response

The four-state machine and PRIVILEGED/STANDARD classification implement a graduated response: full access (TRUSTED), restricted access (SUSPICIOUS), full block (QUARANTINED/REVOKED). Prevents over-reaction to transient issues while protecting high-impact operations.

5.2.4 Complete Four-Dimension Audit Trail

Trust state transitions, security events, collaboration messages, and VC issuance records provide a comprehensive audit trail for forensic investigation and compliance review.

5.2.5 Modular Five-Layer Architecture

The local SQLite VDR can be replaced with a distributed registry; LocalAgentConnector can be replaced with live LLM APIs; trust weights can be reconfigured — all without restructuring the core trust engine or API layers.

5.2.6 W3C Standards Compliance

DID:key and VC implementation follow W3C specifications [1][2], providing a foundation for interoperability with other SSI-compatible systems.

5.3 Limitations

5.3.1 Local VDR — Not Distributed

The VDR is a local SQLite file (vdr.db). It is not tamper-evident, replicated, or accessible across organisations. A production deployment requires a distributed registry.

5.3.2 Structured Simulation — Not Live LLM

LocalAgentConnector produces deterministic capability-specific text responses. It does NOT perform live LLM inference. Real multi-agent scenarios require actual AI endpoint integration.

5.3.3 PARALLEL Workflow Pattern Incomplete

The PARALLEL pattern is schema-supported but not implemented in the execution engine. Only SEQUENTIAL is verified. PARALLEL execution is future scope.

5.3.4 Single-Organisation Scope

All agents must register in the same local VDR. Cross-organisational federated trust is not implemented.

5.3.5 VC API Key Validation Disabled

The API key gate for VC issuance is commented out in vdr/router.py (development mode). Production deployment requires enabling this validation.

5.3.6 Static Trust Weight Configuration

Weights w1=0.6, w2=0.3, w3=0.1 and lambda=0.7 are configured at startup and do not adapt to operational context.

5.4 Potential Applications

5.4.1 Enterprise AI Orchestration

Trust-gated agent collaboration for business process automation; prevents rogue agent actions in multi-step AI pipelines.

5.4.2 Cybersecurity SOC Automation

Continuous monitoring of AI threat hunting agents; restricts compromised agents from executing firewall or isolation actions.

5.4.3 Healthcare AI Agent Networks

Behavioural monitoring and VC-based credential management for agents accessing patient data, supporting HIPAA/GDPR compliance.

5.4.4 Smart Grid and Industrial IoT

Trust scoring for control agents in critical infrastructure; detects anomalies before physical harm from compromised agents.

5.4.5 Decentralised AI Agent Marketplaces

Transparent reputation records and DID-based identity for AI agent service platforms, creating market incentives for reliable behaviour.

5.5 Chapter Summary

TrustChain-AI's principal strengths are its cryptographic identity integration, continuous behavioural monitoring, graduated adaptive authorization, complete audit trail, modular architecture, and W3C standards compliance. Its principal limitations — local VDR, simulation connector, incomplete PARALLEL execution, and static weights — reflect deliberate scope decisions appropriate for a capstone demonstration, and are addressed in future scope.

CHAPTER 6: CONCLUSION AND FUTURE SCOPE

6.1 Project Conclusion

TrustChain-AI successfully demonstrates a decentralised trust intelligence framework for secure autonomous AI agent collaboration. All six project objectives stated in Section 1.3 have been achieved and independently verified through a test suite of 77 automated tests (77/77 PASSED, Python 3.12.3, 2026-10-16).

DID:key Identity (Objective 1)

The DIDKey class (akta/did.py, 452 lines) implements the W3C DID:key method with Ed25519 keypair generation, multicodec prefix encoding (0xED01), base58 multibase construction, DID Document generation, and Ed25519 signature verification. Verified in test_e2e_lifecycle and test_cli.

Verifiable Credentials (Objective 2)

The VerifiableCredential class implements the W3C VC Data Model v1.1 with Ed25519Signature2020 proofs and URDNA2015 JSON-LD canonicalisation. VCs are stored in, and retrieved from, the local SQLite VDR.

Temporal EMA Trust Scoring (Objective 3)

The trust engine evaluates S_i, C_i, L_i to compute P_i(t), applies EMA with lambda=0.7 to compute T_i(t), and transitions agents through TRUSTED/SUSPICIOUS/QUARANTINED/REVOKED states. End-to-end tests verify degradation, quarantine, recovery, and revocation paths.

Adaptive Authorization (Objective 4)

The AdaptivePolicyEnforcer (211 lines) classifies operations as PRIVILEGED or STANDARD and enforces trust-state-dependent access control. All 8 trust state × operation combinations are test-verified.

Trust-Weighted Discovery (Objective 5)

The discovery engine filters by capability and trust state, then ranks by composite match score (0.4 trust + 0.3 reputation + 0.2 success + 0.1 latency). Verified in test_feature3_trust_aware_agent_matching.

Sequential Workflow Orchestration (Objective 6)

The execution engine orchestrates sequential pipelines with output-to-input data chaining, per-step trust verification, and complete node-level state recording. Verified in test_feature2_multi_agent_workflow_orchestrator.

6.2 Contribution Summary

6.2.1 Integrated Identity-Trust Framework

First demonstration of a single integrated and testable system combining W3C DID:key cryptographic identity, W3C Verifiable Credentials with Ed25519Signature2020, and a formal Exponential Moving Average behavioural trust scoring engine.

6.2.2 Adaptive Policy Enforcement for AI Agents

The AdaptivePolicyEnforcer operationalises NIST Zero Trust Architecture principles [15] specifically for the AI agent domain, introducing an operation classification taxonomy and trust-state-dependent enforcement mechanism for AI collaboration contexts.

6.2.3 Trust-Aware Multi-Factor Agent Matching

The agent discovery formula incorporates four factors — trust score, peer reputation, task success rate, and response latency — into a composite match score that ensures trust-aware agent selection for collaborative tasks.

6.3 Future Scope

6.3.1 Distributed Verifiable Data Registry

Replace SQLite with a distributed registry (Hyperledger Indy, Ethereum-based DID registry) for tamper-evidence, replication, and cross-organisational accessibility.

6.3.2 Live LLM Agent Integration

Extend LocalAgentConnector with real LLM API adapters (OpenAI GPT, Google Gemini, Anthropic Claude) for genuine AI agent task execution.

6.3.3 PARALLEL Workflow Execution

Implement concurrent node execution using asyncio task coordination for PARALLEL workflow pattern, enabling simultaneous multi-capability task pipelines.

6.3.4 Federated Trust Across Organisations

A trust federation protocol allowing agents from different organisations' VDRs to establish mutual trust relationships via VC-based trust attestation.

6.3.5 Dynamic Trust Weight Adaptation

Context-sensitive weight adjustment: higher w2 (compliance) for safety-critical operations; machine-learning-based weight optimisation from historical trust accuracy.

6.3.6 W3C VC Revocation Registry

Implement StatusList2021 for scalable credential revocation, replacing the current deletion-based approach.

6.3.7 Zero-Knowledge Proof Credential Presentation

Enable agents to prove credential attributes (e.g., trust level above SUSPICIOUS) without revealing full credential content, using BBS+ signature scheme.

6.4 Final Remarks

TrustChain-AI establishes a solid, testable foundation for trust-aware multi-agent AI systems. The implementation demonstrates that cryptographic identity, continuous behavioural trust scoring, and adaptive authorization can be integrated into a coherent, modular, standards-compliant framework that is verifiable through automated testing. As AI agent deployment scales across enterprise, healthcare, security, and infrastructure domains, the problems addressed by TrustChain-AI — identity, trust, and authorization — will become central to responsible AI systems design. The core trust intelligence model constitutes a reusable, independently testable contribution to the growing field of AI agent governance.

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APPENDIX A: API ENDPOINT REFERENCE

Base URL: http://localhost:8050. All endpoints except /health require JWT Bearer token.

Method	Path	Role	Description
POST	/auth/register	None	Register user
POST	/auth/login	None	Authenticate; returns JWT pair
POST	/auth/refresh	None	Rotate refresh token
POST	/auth/logout	Any	Revoke refresh token
GET	/auth/me	Any	Current user profile
GET	/auth/users	ADMIN	List all users
POST	/auth/users	ADMIN	Create user
PUT	/auth/users/{id}	ADMIN	Update user role/status
GET	/auth/audit-events	ADMIN/OP	Security audit events
POST	/registry/agents	ADMIN/OP	Register agent
GET	/registry/agents	Any	List agents
GET	/registry/agents/{id}	Any	Agent detail
PUT	/registry/agents/{id}	ADMIN/OP	Update agent
DELETE	/registry/agents/{id}	ADMIN	Remove agent
POST	/registry/agents/{id}/telemetry	Any	Submit telemetry → trust recompute
GET	/registry/agents/{id}/trust	Any	Current trust score/state
GET	/registry/agents/{id}/trust-audit	Any	Trust audit trail
POST	/registry/agents/{id}/revoke	ADMIN	Admin revocation
POST	/registry/discover	Any	Discovery + ranked matching

POST	/registry/authorize	Any	Check adaptive authorization
POST	/registry/collaborate	ADMIN/OP	Initiate collaboration
GET	/registry/collaborations/{id}	Any	Session details
POST	/registry/collaborations/{id}/message	Any	Send message
POST	/registry/collaborations/{id}/complete	Any	Complete session
POST	/registry/workflows	ADMIN/OP	Create workflow
GET	/registry/workflows/{id}	Any	Workflow status
POST	/registry/workflows/{id}/execute	ADMIN/OP	Execute workflow
GET	/registry/agents/{id}/ratings	Any	Agent ratings
POST	/registry/agents/{id}/rate	Any	Submit rating
POST	/vdr/credentials	ADMIN/OP	Issue VC
GET	/vdr/credentials	Any	List VCs
GET	/vdr/credentials/{id}	Any	Get VC
DELETE	/vdr/credentials/{id}	ADMIN	Revoke VC
POST	/vdr/did/generate	Any	Generate DID:key
POST	/vdr/did/resolve	Any	Resolve DID Document
POST	/vdr/did/verify	Any	Verify Ed25519 signature
GET	/health	None	System health probe
GET	/	None	Root endpoint

APPENDIX B: DATABASE SCHEMA REFERENCE

Local SQLite VDR (vdr.db), WAL mode, SQLAlchemy 2.0. Full schema in docs/blackbook/evidence/db_schema.md.

Table Name	Key Columns
agent_trust_states	agent_did (UK), temporal_trust_score (FLOAT), trust_state (VARCHAR), last_updated
telemetry_logs	agent_did, success (BOOL), compliance_event (BOOL), latency_ms (FLOAT), timestamp
trust_audit_trails	agent_did, previous_state, new_state, previous_score, new_score, reason, triggered_by
stored_vcs	vc_id (PK), issuer_did, subject_did, credential_type, credential_data (TEXT), issued_at
agents	agent_id (PK), did (UK), name, capabilities (JSON), status, health_status, reputation_score
agent_ratings	rating_id (PK), agent_id (FK), rater_did, rating (1.0-5.0), feedback
collaboration_sessions	session_id (PK), requester_did, target_did, task_description, status, task_result
collaboration_events	event_id (PK), session_id (FK), sender_did, message_type, content, timestamp
workflows	workflow_id (PK), name, pattern (SEQUENTIAL/PARALLEL), status, result
workflow_nodes	node_id (PK), workflow_id (FK), agent_did, required_capability, step_order, output_data
users	user_id (PK), username (UK), role, password_hash (Argon2id), failed_login_attempts, locked_until
refresh_tokens	id (PK), jti (UK), token_hash, user_id (FK), expires_at, revoked_at
security_audit_events	event_id (PK), event_type, user_id (FK), ip_address, timestamp, details (JSON)

APPENDIX C: TEST RESULTS SUMMARY

Appendix C — Test Results (full output in evidence/test_results.txt)

```
Test Session: 2026-10-16
Platform: Windows 11, Python 3.12.3
Framework: pytest
Duration: 11.37 seconds
Result: 77 passed, 0 failed, 0 errors, 216 warnings (Pydantic v2
deprecations)

Key passing tests:
  test_e2e_full_agent_lifecycle_success_path                    PASSED
  test_e2e_telemetry_degradation_and_quarantine_path            PASSED
  test_e2e_recovery_and_administrative_revocation_path          PASSED
  test_multi_agent_registry_and_discovery                       PASSED
  test_collaboration_session_lifecycle_and_telemetry            PASSED
  test_quarantined_agent_collaboration_rejection                PASSED
  test_feature1_agent_health_monitoring                         PASSED
  test_feature2_multi_agent_workflow_orchestrator               PASSED
  test_feature3_trust_aware_agent_matching                      PASSED
  test_system_health_healthy                                    PASSED
  test_system_health_database_failure_simulation                PASSED
  test_system_health_trust_engine_failure_simulation            PASSED
  test_system_health_simultaneous_failure                       PASSED
  (+ 64 further tests across authentication, adaptive policy, CLI)

============================== 77 passed, 216 warnings in 11.37s
==============================

Full output: docs/blackbook/evidence/test_results.txt
```

APPENDIX D: TRUST MATHEMATICS REFERENCE

D.1 Symbol Definitions

Symbol	Definition	Range
P_i(t)	Multi-factor performance score	[0.0, 1.0]
T_i(t)	Temporal trust score (EMA)	[0.0, 1.0]
S_i(t)	Task success rate (last n events)	[0.0, 1.0]
C_i(t)	Policy compliance rate (last n events)	[0.0, 1.0]
L_i(t)	Normalised latency = min(avg_ms/2000, 1)	[0.0, 1.0]
w1=0.6, w2=0.3, w3=0.1	Trust factor weights (sum=1.0)	Fixed
lambda=0.7	EMA decay factor	Fixed
n=10	Evaluation window size	Fixed
tau_T=0.8	TRUSTED state threshold	Fixed
tau_S=0.5	SUSPICIOUS state threshold	Fixed

D.2 Core Equations

L_i(t) = min(avg_latency_ms / 2000, 1.0)

P_i(t) = 0.6*S_i(t) + 0.3*C_i(t) + 0.1*(1 - L_i(t)) ... (Eq. 1)

T_i(t) = 0.7*P_i(t) + 0.3*T_i(t-1) ... (Eq. 2)

match_score_i = 0.4*T_i + 0.3*(r_i/5.0) + 0.2*sigma_i + 0.1*(1-lambda_i) ... (Eq. 3)

D.3 Boundary Conditions

Initial trust: T_i(0) = 1.0 (TRUSTED). REVOKED state: terminal; telemetry does not update score. Empty window: T_i(t) = 1.0 preserved. Weight integrity: w1+w2+w3 = 1.0, lambda+(1-lambda) = 1.0.

APPENDIX E: ALGORITHM PSEUDOCODES

E.1 Verifiable Credential Issuance

Algorithm 3 — VC Issuance (akta/credentials.py)

```
Input:  issuer_did, issuer_private_key, subject_did, credential_claims
Output: vc_document (JSON-LD with Ed25519Signature2020 proof)

1.  vc ← build_vc_structure(issuer_did, subject_did, credential_claims)
    // Includes @context, type, issuer, issuanceDate, credentialSubject
2.  canonical ← urdna2015_canonicalize(vc)  // via PyLD
3.  signature ← ed25519_sign(canonical, issuer_private_key)  // via PyNaCl
4.  proof_value ← 'z' + base58btc(signature)  // multibase
5.  vc['proof'] ← {
      'type': 'Ed25519Signature2020',
      'created': now_iso8601(),
      'verificationMethod': issuer_did + '#' + multibase_key,
      'proofPurpose': 'assertionMethod',
      'proofValue': proof_value
    }
6.  INSERT vc into stored_vcs
7.  return vc
```

E.2 Adaptive Policy Enforcement

Algorithm 4 — Adaptive Policy Enforcement (akta/adaptive_auth.py)

```
Input:  agent_did, operation_type
Output: decision (PERMIT / DENY), reason

1.  trust_state ← get_trust_state(agent_did)  // from agent_trust_states
2.  if trust_state == REVOKED:
        DENY; reason = 'Agent permanently revoked'; INSERT audit_event;
return
3.  if trust_state == QUARANTINED:
        DENY; reason = 'Agent quarantined'; INSERT audit_event; return
4.  if operation_type in PRIVILEGED_OPERATIONS:
        if trust_state == SUSPICIOUS:
            DENY; reason = 'Privileged op denied to SUSPICIOUS'; INSERT
audit_event; return
5.  // TRUSTED or SUSPICIOUS + STANDARD
    INSERT audit_event (if SUSPICIOUS)
    PERMIT; return
```

APPENDIX F: TECHNOLOGY STACK DETAILS

All versions verified from pip list and package.json on 2026-10-16.

Appendix F — Complete Dependency Versions

```
# Backend (Python 3.12.3 virtualenv)
FastAPI==0.141.1        Uvicorn==0.52.4          SQLAlchemy==2.0.52
Pydantic==2.13.5        PyNaCl==1.6.2            PyJWT==2.14.0
PyLD==3.2.0             passlib==1.7.4           argon2-cffi==25.1.0
base58==2.1.1           httpx==0.28.1            opentelemetry-api==1.44.0
opentelemetry-sdk==1.44.0  python-dotenv==1.2.3

# Frontend (Node.js, package.json)
react==19.2.8           react-dom==19.2.8        vite==8.2.2
react-router-dom==7.18.3  recharts==3.10.1       tailwindcss==4.3.3
lucide-react==1.38.0    axios==1.20.0
```

APPENDIX G: SELECTED SOURCE CODE EXCERPTS

G.1 Trust Score Computation (akta/vdr/trust_crud.py)

G.1 — Trust Score Computation (simplified excerpt, akta/vdr/trust_crud.py)

```
def compute_trust_score(db, agent_did: str, new_event: dict) -> dict:
    # Insert new telemetry event
    db.add(TelemetryLog(agent_did=agent_did, **new_event))
    db.commit()

    # Load last 10 events
    events = (db.query(TelemetryLog)
                .filter(TelemetryLog.agent_did == agent_did)
                .order_by(TelemetryLog.timestamp.desc())
                .limit(10).all())
    n = len(events)

    S_i = sum(1 for e in events if e.success) / n
    C_i = sum(1 for e in events if e.compliance_event) / n
    avg_l = sum(e.latency_ms for e in events) / n
    L_i  = min(avg_l / 2000.0, 1.0)
    P_i  = 0.6 * S_i + 0.3 * C_i + 0.1 * (1 - L_i)

    state_rec =
db.query(AgentTrustState).filter_by(agent_did=agent_did).first()
    T_prev = state_rec.temporal_trust_score if state_rec else 1.0
    T_i    = 0.7 * P_i + 0.3 * T_prev

    if   T_i >= 0.8: trust_state = TrustState.TRUSTED
    elif T_i >= 0.5: trust_state = TrustState.SUSPICIOUS
    else:            trust_state = TrustState.QUARANTINED

    # ... update DB and write audit trail ...
    return {"trust_score": T_i, "trust_state": trust_state.value}
```

G.2 DID:key Generation (akta/did.py)

G.2 — DID:key Generation (simplified excerpt, akta/did.py)

```
import nacl.signing, base58

MULTICODEC_ED25519 = bytes([0xED, 0x01])

def generate_did_key():
    signing_key  = nacl.signing.SigningKey.generate()
    public_key   = bytes(signing_key.verify_key)
    prefixed     = MULTICODEC_ED25519 + public_key
    multibase    = 'z' + base58.b58encode(prefixed).decode()
    did_uri      = f'did:key:{multibase}'
    vm_id        = f'{did_uri}#{multibase}'
    did_document = {
        '@context': ['https://www.w3.org/ns/did/v1',
                     'https://w3id.org/security/suites/ed25519-2020/v1'],
        'id': did_uri,
        'verificationMethod': [{
            'id': vm_id, 'type': 'Ed25519VerificationKey2020',
            'controller': did_uri, 'publicKeyMultibase': multibase
        }],
```

```
    'authentication': [vm_id], 'assertionMethod': [vm_id],
}
return did_uri, did_document, bytes(signing_key)
```

APPENDIX H: FINAL QA CHECKLIST

Use before submission. Tick each item when verified.

Preliminary Pages

☐ Title page: project title, degree, department, university, student names, SAP IDs, mentor, year

☐ Certificate: signed by internal guide, HoD, external examiner; seal space present

☐ Acknowledgement: mentor, department, university acknowledged

☐ Abstract: 250-300 words; problem, solution, methodology, results, contribution

☐ TOC, List of Figures, List of Tables, Abbreviations, Symbols all present

Trust Model Accuracy (CRITICAL)

☐ P_i(t) = 0.6*S + 0.3*C + 0.1*(1-L) — EXACT weights w1=0.6, w2=0.3, w3=0.1

☐ T_i(t) = 0.7*P + 0.3*T_prev — EXACT lambda=0.7

☐ TRUSTED: T >= 0.8; SUSPICIOUS: 0.5 <= T < 0.8; QUARANTINED: T < 0.5

☐ REVOKED: terminal, admin-only, not threshold-triggered

☐ Initial trust: 1.0 (TRUSTED); window: 10 events

Honest Limitation Statements

☐ VDR is local SQLite (vdr.db) — NOT distributed or blockchain — stated clearly

☐ LocalAgentConnector is structured local execution engine — NOT live LLM — stated clearly

☐ PARALLEL workflow: schema-only, not tested — labeled as future scope

☐ VC API key validation disabled in development mode

Test Results

☐ 77 tests PASSED, 0 failed — stated in Ch.4 and Appendix C

☐ No fabricated test results

References

☐ All 15 references are real, verifiable papers/standards

☐ In-text citations [1]-[15] match reference list

☐ IEEE formatting used consistently

Document Formatting

☐ Times New Roman 12pt body

☐ 1.5 line spacing

☐ Left margin 1.5 in, other margins 1 in

☐ Chapter titles 18pt bold; section 14pt bold; subsection 12pt bold

☐ Figure captions 10pt below; table captions 10pt above

☐ Student name/roll/SAP filled (replace all placeholders)